

AI Models Need An Abstraction Like Printers Need A Driver

Why binding applications to one AI vendor is the new printer lock-in, and why model abstraction is the strategic fix.

Model abstraction matters more than being multi-vendor on the supply chain. The **Proxy of Record** is the printer driver: security, manageability, transparency, and accountability on the request path. Multi-vendor is what that abstraction makes safe when the best lane for the job lives elsewhere.

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Why talk about printer drivers in an AI article. Because applications do not develop toward a specific printer, no matter how good that printer is. The driver is the abstraction that keeps a good product from painting itself into a corner when a better device shows up. Model abstraction is the same move. It matters more than being multi-vendor on supply chain, infrastructure, or headcount. Without it, every agent and workflow hard-binds to one vendor's quirks. When the leaderboard moves, or the tariff shape changes, or a cheaper lane appears for a mundane job, the product is stuck.

Once that abstraction is in place, three pressures still break teams that skip it: price gaps that are orders of magnitude, the habit of putting the wrong model on the wrong job, and the lock that turns a volume commit into a painted corner.

Price Is Orders of Magnitude Different

Price and performance gaps are not a twenty-five percent shopping delta. They are orders of magnitude.

List prices accessed 2026-09-08 make the point without a sales deck. Anthropic Claude Fable 5.1 output lists at \$50 per million tokens, which is \$0.00005 per token or \$0.05 per thousand tokens. Z.AI GLM-4.6V-FlashX input lists at \$0.04 per million tokens, which is \$0.00000004 per token or \$0.00004 per thousand tokens. Same unit. About 1,250 times apart. Those SKUs are not the same job. That is the point.

Across vendors the best price and performance for a discrete job often simply does not live in your primary shop. Coding versus internet research. Creating an image versus OCR. Scientific formulas versus summarizing text. Why do people with resources refuse to send every child to the one same university. Because no university is the best price and performance for every field of study. Actors do not study at Harvard. Why pay that price if they do not want to be an attorney or a doctor.

Don't Use the Wrong Model for the Wrong Job

The recurring mistake is using the newest frontier model for summarizing email, translating documents, writing routine code, or assembling workflows. Summarizing email is discrete. Everything you need is in the text. Writing an industry analysis is a different job. If you use the same model for both, you waste money.

Right model for the right job means different lanes for different work. Multi-model inside one shop can help. It is still one commercial calendar and one outage domain. Sometimes the cheap-enough lane for the discrete job lives elsewhere. Sometimes the frontier lane for the hard job does. The decision is the job, not brand loyalty.

Why The Driver Analogy Works

Multi-vendor without abstraction is just more keys. The driver is what keeps the product from locking to one SKU family when the leaderboard moves. Model abstraction is the strategic layer. Multi-vendor is what that layer makes safe when the best lane for the job lives elsewhere.

Telecom already solved the metered version of this problem. Wholesale voice and enterprises with multiple SIP trunks do not hard-commit every minute to one upstream carrier. They load rate decks from several carriers into a routing engine. Each call ranks qualifying routes by cost and quality floors, attempts the primary path, fails over when a route dies, and logs the actual route on a call detail record. That is multi-sourcing with a chokepoint. Not chaos. AI mix-by-use-case is the same pattern with models instead of trunks and a Token Detail Record instead of a call detail record. Least-cost routing without accurate rates and quality gates produces bad calls and bad bills. Multi-vendor AI without a controlled gateway and a meter produces the same failure mode.

Distributing first-party API keys to every employee across vendors is messy to administer. A centralized proxy that binds a user key on the outside to account keys on the vendor side is cleaner. Disable one proxy key and you disable that user's access across vendors. Shadow first-party keys beside the proxy leave the meter. Call that chokepoint a Proxy of Record. One place to mint user access, meter every call, and cut a user off across vendors. Keys leave laptops. Admin stays central. Every request leaves a timesheet.

Don't Get Painted Into a Corner

Binding to a native stack only makes sense in a single-discipline shop with a stable model landscape. That landscape is not stable. New vendors, models, tools, and leaderboards arrive continuously. It moves too fast for painting a company to one vendor as lasting architecture. Single-focus shops that only need AI for one thing are rare. If a CEO wants to say the company only needs this tool for one thing, let them. They have a right to that position.

Volume buying, take-or-pay, and breakage can still be an honest single-vendor win when a coding shop has tested the models and wants a year of committed volume with one frontier vendor to drive down the per-token rate. That case can be true. Facts still make it rare. What fails after the commit is the tribal mindset. We cannot use OpenAI because we are an Anthropic shop. We wish we had that model but DeepSeek is Chinese and we do not use Chinese models. Asking for the ratio of births to deaths in the United States over fifty years is public information. Why would it matter if you sent that job to a much cheaper lane as long as you watch the context the question may carry.

Buying more vendors is not a mandate to put every task through a model. Email triage, OCR, and industry analysis are different jobs with different economics. Some work should stay human or stay on a cheap specialized lane. Abstraction lets you choose. Stay unbound enough that a better model, or a cheaper lane for a mundane job, does not require rewriting the product. Do not paint yourself into a corner.

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